Information Fusion 124 (2025) 103404



Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus



Full length article



QD-MSA : A quantum distributed tensor network framework for multimodal sentiment analysis

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ARTICLE INFO

Keywords:
Sentiment analysis
Multimodal sentiment analysis
Quantum computing
Distributed quantum computing
Quantum machine learning

ABSTRACT

Multimodal sentiment analysis, which integrates data types such as audio, text, and images, is increasingly vital for understanding emotional content in the era of social media and short video platforms. Quantum computing, with its inherent characteristics like superposition and entanglement, is conceptually well-suited for multimodal learning, particularly for modal fusion. However, current quantum computers face limitations, such as a restricted number of usable qubits, hindering their ability to surpass classical computing (quantum supremacy). In this work, we propose QD-MSA, a quantum distributed multimodal sentiment analysis framework, which is the first to apply quantum circuit splitting techniques to multimodal sentiment analysis, reducing qubit usage from n to $n/2+1$. This advancement enables the execution of more complex quantum programs on Noisy Intermediate-Scale Quantum (NISQ) devices by partially overcoming qubit scarcity. Additionally, QD-MSA contains a novel workflow that integrates our model into quantum computer clusters, significantly enhancing computational performance and unlocking the potential of NISQ-era quantum computers. By combining classical neural networks for feature extraction with quantum models for feature fusion, our approach conserves quantum resources while achieving superior performance. Experimental evaluations on the CMU-MOSEI and CMU-MOSI datasets demonstrate that our model achieves comparable or superior performance to deep learning-based methods, with notable improvements in key metrics. Furthermore, our work represents the first successful integration of quantum computing principles into multimodal sentiment analysis, with experiments confirming that the proposed model significantly outperforms classical approaches relying solely on quantum-inspired strategies. These contributions establish a scalable and efficient framework for multimodal sentiment analysis, leveraging both classical and quantum computing paradigms to advance the field.

1. Introduction

With the rapid development of personal social media and short video platforms, an enormous volume of video data has emerged on the internet, leading individuals to increasingly utilize videos as a medium for self-expression. Consequently, sentiment analysis of multimodal data has become a significant research topic in the academic community. Unlike traditional sentiment analysis, which typically extracts emotional content solely from textual data, multimodal sentiment analysis involves extracting and analyzing features from multiple data modalities such as audio, text, images, and even EEG signals. This added complexity necessitates more sophisticated approaches, generally divided into two primary stages: feature extraction and feature fusion.

Currently, deep learning methods have become the mainstream approach [1–3] for addressing multimodal sentiment analysis problems. Models such as the Tensor Fusion Network (TFN) [4] and the Multi-Attention Recurrent Network (MARN) [5] have demonstrated state-of-the-art results in this domain. However, as data complexity grows, their parameter counts and computational overheads escalate sharply, challenging scalability and efficiency. For example, TFN computes cross-modal correlations via tensor outer products, causing the feature tensor dimension to scale cubically. In a typical video classification scenario with 1024-dimensional visual features, 300-dimensional audio features, and 300-dimensional text features, the fused tensor contains on the order of $1024\times300\times300 \approx 9.22\times10^7$ elements. This leads to model sizes between 80 and 200 MB (approximately 25–60 million parameters), and training TFN on a dataset like CMU-MOSI requires

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https://doi.org/10.1016/j.inffus.2025.103404
Received 21 January 2025; Received in revised form 3 June 2025; Accepted 4 June 2025
Available online 20 June 2025
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roughly 10–12 h on a single NVIDIA V100 GPU with over 16 GB of GPU memory. MARN, which employs multi-level attention mechanisms and LSTM units—including a dedicated cross-modal memory cell—incurs even higher costs: it has about 1.2 times the parameter count of TFN (approximately 30–70 million parameters) and can demand upwards of 20 GB of GPU memory during training. These significant resource requirements motivate the investigation of alternative, more lightweight computational paradigms.

In recent years, the capabilities of quantum computers [6–8] have surged, demonstrating potential advantages over classical computers in specific problem domains. The fundamental concepts of superposition and entanglement in quantum computing exhibit correlations with feature extraction and feature fusion processes in multimodal sentiment analysis. Leveraging the intrinsic strengths of quantum computing presents a viable pathway for enhancing multimodal sentiment analysis. Existing works have begun to integrate quantum computing concepts into classical models, yielding promising results. For instance, the Quantum Fuzzy Neural Network (QFNN) [9] applies quantum and fuzzy logic principles to improve sentiment and sarcasm detection, while the framework proposed in [10] utilizes quantum theoretical principles to construct multimodal representations for video sequences, thereby enhancing the accuracy of sentiment analysis based on textual queries. Additionally, some studies have employed genuine quantum models, such as the Quantum Non-negative Matrix Factorization (QNMF) system [11], which utilizes a Quantum Convolutional Neural Network (QCNN) and a Variational Quantum Classifier (VQC) to efficiently process and fuse multimodal medical data from the Internet of Medical Things (IoMT).

Despite significant advances in quantum computing, devices in the Noisy Intermediate-Scale Quantum (NISQ) era continue to face severe limitations in terms of available qubits and noise resilience. In this work, we introduce a novel quantum-hybrid model and a distributed workflow that together enable practical applications of quantum computing for analyzing video data and classifying human sentiment. By leveraging quantum circuit splitting techniques, our approach decomposes a quantum circuit requiring n qubits into several sub-circuits each needing only $\frac{n}{2}+1$ qubits, saving almost 50% of qubit usage. This not only extends the capabilities of limited-qubit quantum computers but also allows for the execution of larger, more complex quantum programs. Furthermore, our model integrates classical neural networks for robust feature extraction with quantum models for high-dimensional pattern recognition, achieving performance that rivals or exceeds state-of-the-art deep learning methods while utilizing only 1% of the parameters typically required by classical models in modal fusion.

The contributions of this work are fivefold:

1. We propose a novel distributed workflow that integrates our quantum-hybrid model into clusters of quantum computers, significantly enhancing computational performance and unlocking the full potential of NISQ devices for complex multimodal sentiment analysis.
2. We introduce a pioneering quantum multimodal sentiment analysis model that employs quantum encoding and quantum tensor networks to fuse diverse modalities. This contribution not only highlights a new application of quantum computing in the NISQ era but also underscores the importance of our approach in practical video analysis and human sentiment classification.
3. Our method achieves effective fusion of multiple modalities with remarkable parameter efficiency, requiring only 1% of the parameters used by conventional deep learning-based models in modal fusion, thereby enhancing both efficiency and scalability.
4. We apply quantum circuit splitting techniques to the problem of multimodal sentiment analysis for the first time, reducing qubit usage from n to $\frac{n}{2}+1$, saving almost 50% of qubits. This significant reduction enables the execution of larger quantum circuits on devices with limited qubits.
5. Extensive experiments on the CMU-MOSEI and CMU-MOSI datasets demonstrate superior performance, with accuracies of 81.62% and 77.80% respectively, outperforming both state-of-the-art deep learning models and classical models that utilize quantum-inspired strategies.

The implementation for our model is available at https://github.com/XDU-AI-LYYLab/QD-MSA.

2. Related work

2.1. Quantum computation theory in a nutshell

2.1.1. Quantum computation theory

Quantum computation [12] leverages the fundamental principles of quantum mechanics [13] to transform the paradigm of information processing. Central to this revolutionary approach is the quantum bit, commonly known as the qubit, which differs significantly from classical bits. Unlike classical bits restricted to binary states (0 or 1), qubits exploit quantum superposition, allowing them to exist simultaneously in multiple states. Mathematically, a qubit state [14] is represented as:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Here, $|\psi\rangle$ denotes the state vector of a qubit, while $|0\rangle$ and $|1\rangle$ are basis states. The complex coefficients a and b, known as amplitudes [15], satisfy the normalization condition $|a|^2 + |b|^2 = 1$, determining the probabilities of the qubit collapsing into each basis state upon measurement. This intrinsic feature of quantum superposition facilitates quantum parallelism in computational processes.

For multi-qubit systems [16], the overall state is expressed through the tensor product of individual qubit states, resulting in exponential growth [17] of the system's state space:

$$|\psi_1\rangle \otimes |\psi_2\rangle = |\psi_1\psi_2\rangle$$

Quantum logic gates [18], fundamental to quantum computation, manipulate qubit states via unitary transformations [19]. These transformations preserve quantum coherence and maintain reversibility, essential properties for quantum operations. A quantum gate applies a unitary operator U to a qubit state $|\psi\rangle$, transforming it into a new state $|\psi'\rangle$:

$$U|\psi\rangle = |\psi'\rangle$$

The operator U satisfies the unitary condition $UU^\dagger = U^\dagger U = I$, where I represents the identity matrix. This enables simultaneous manipulation of multiple qubits and supports computational parallelism through quantum superposition.

Measurement in quantum mechanics, critical for obtaining classical information from quantum states [20], is inherently probabilistic. Upon measurement, a quantum state $|\psi\rangle$ collapses into one of its basis states associated with an eigenvalue λ. The probability of observing this outcome is given by:

$$\Pr(\lambda) = \|\Pi_\lambda|\psi\rangle\|^2$$

Here, Π_λ represents the projection operator onto the eigenspace corresponding to eigenvalue λ. The probabilistic nature of quantum measurements contrasts starkly with deterministic outcomes observed in classical computation.

Quantum computing demonstrates remarkable potential due to its inherent alignment with quantum phenomena and its parallel processing capabilities. The structural similarity between quantum computers and quantum systems permits efficient simulation of natural quantum phenomena, vastly surpassing classical computing capabilities [21,22]. This advantage is particularly pronounced in areas such as quantum chemistry and material science, where classical simulations encounter exponential complexity. Quantum parallelism, allowing simultaneous exploration of multiple computational pathways, provides exponential

speedup for specific problems. While classical computation demands exponentially increasing resources (c^n bits for n quantum components), quantum computation scales linearly ($k \times n$ qubits) for similar simulations, illustrating its substantial efficiency. Despite these advantages, significant challenges persist, notably in extracting meaningful classical information from quantum simulations, as measurement inherently collapses quantum states into limited classical data. Nonetheless, quantum computing constitutes a fundamentally superior computational framework, capable of addressing problems intractable by classical supercomputers and fostering advancements in quantum technology and fundamental quantum research.

2.1.2. Quantum Machine Learning in NISQ era

Quantum Machine Learning (QML) emerges as an interdisciplinary field integrating quantum computation with classical machine learning paradigms [23]. The core objective of QML is to leverage distinct quantum mechanical principles, notably superposition and entanglement [24], to potentially achieve advancements in computational tasks such as data classification, pattern recognition, and complex optimization problems. By exploiting the capacity of quantum systems to navigate exponentially large Hilbert spaces, QML offers novel computational pathways. This potential for fundamentally different information processing capabilities positions QML as a compelling and potentially powerful approach towards the ambitious goal of Artificial General Intelligence (AGI), exploring computational capabilities beyond classical algorithms.

Current research and application development in QML are predominantly situated within the operational limitations of the Noisy Intermediate-Scale Quantum (NISQ) era [25,26]. This period is defined by the availability of quantum processors [27,28] that feature a modest number of qubits (typically ranging from 50 to several hundred) and are significantly hampered by inherent noise, gate errors, and decoherence, lacking the requisite scale and fidelity for comprehensive quantum error correction. For instance, current superconducting qubits exhibit coherence times on the order of 50–100 μs, while single and two-qubit gate error rates often range from 0.1% to 1% [28,29], imposing strict limits on circuit depth and overall execution time. Such decoherence and error accumulation can severely degrade algorithmic performance in real-world applications, necessitating sophisticated error mitigation, dynamic calibration procedures, and hybrid quantum–classical workflows to preserve fidelity. Concurrently, the prospect of future fault-tolerant quantum computers looms large, carrying profound implications; most notably, algorithms like Shor's [30] possess the theoretical capability to break widely deployed public-key cryptographic systems such as RSA and ECC in polynomial time, posing a significant security threat [31–33]. This looming vulnerability has catalyzed intensive research and standardization efforts in post-quantum cryptography (PQC) [34–36], aiming to establish cryptographic protocols secure against quantum attacks and heralding a new era in cybersecurity [37,38]. The immediate constraints imposed by NISQ technology, alongside the transformative potential highlighted by applications like Shor's algorithm, underscore the critical need for developing specialized quantum algorithms, such as VQAs, that can operate effectively on current, albeit imperfect, hardware while providing a roadmap for eventual deployment in more robust, error-corrected quantum environments [29].

Within this challenging yet dynamic context, Variational Quantum Algorithms (VQAs) [39] have become a prominent and highly competitive class of algorithms tailored specifically for NISQ devices [40]. VQAs utilize a hybrid quantum–classical computational model [41]. In this model, a Parameterized Quantum Circuit (PQC) [42], characterized by adjustable parameters, is executed on the quantum processor. Measurement outcomes from the quantum execution are then relayed to a classical optimizer, which iteratively refines the circuit parameters to minimize a predefined cost function, thereby steering the computation towards a solution for the target problem [43]. This hybrid strategy is particularly well-suited for the NISQ setting due to its potential to yield meaningful results even with shallow circuits and its relative robustness against hardware noise compared to algorithms requiring deep, fault-tolerant circuits [44,45]. Consequently, VQAs represent a leading strategy for exploring and implementing QML tasks on present-day quantum computers, serving as a crucial bridge towards the eventual realization of fault-tolerant quantum computation [46–48]. Despite ongoing challenges, including the optimization difficulties posed by barren plateaus [49] and the inherent sensitivity to noise [50,51], VQAs remain a vital focus of research for harnessing near-term quantum computational capabilities [52–55].

2.2. Multimodal sentiment analysis

2.2.1. Multimodal sentiment analysis fundamentals

Sentiment analysis has become increasingly significant in the age of social media, where users often express emotions and opinions through a combination of text, audio, and visual content. This surge in multimodal communication has led to the development of Multimodal Sentiment Analysis (MSA) [56], which integrates different data forms, such as text, audio, and visual inputs, to provide a more accurate and comprehensive understanding of sentiment compared to unimodal sentiment analysis [57–59]. MSA is based on the insight that sentiments are frequently conveyed through multiple channels simultaneously, and the effective fusion of these modalities leads to a richer interpretation of the underlying emotions.

Multimodal Sentiment Analysis (MSA) can be formally defined [60] as a composite function that synthesizes and evaluates various input modalities to determine the sentiment or emotional content being expressed. This function incorporates data from text, audio, and visual channels, leveraging the unique strengths of each modality to offer a nuanced and contextually enriched interpretation of sentiment. Mathematically, the function can be represented as follows:

$$MSA : (T, A, V) \rightarrow S \tag{1}$$

Here, T represents the textual data input, A denotes the auditory data input, V indicates the visual data input, and S is the resulting sentiment, which can be classified into discrete categories (such as positive, negative, or neutral) or quantified as a continuous variable that reflects the intensity of the sentiment.

This multi-modal approach assumes that the convergence of diverse communicative cues—linguistic, acoustic, and visual—provides a more robust framework for sentiment analysis. By combining these different inputs, MSA is better equipped to capture subtle nuances and resolve contradictions in emotional expressions that might remain unnoticed in unimodal analyses. The effectiveness of this approach is especially evident in contexts where verbal communication is supplemented or contradicted by non-verbal cues, such as facial expressions or vocal intonations, making it essential to adopt a holistic analytical method to accurately interpret the expressed sentiment.

2.2.2. Quantum circuit cutting method

Quantum circuit cutting is an important approach within quantum circuit decomposition [61], facilitating the division of large quantum circuits into smaller, manageable subcircuits suitable for execution on current quantum hardware. The cutting methods employed in our proposed framework are based on decomposing quantum circuits into linear combinations of unitary operations using orthonormal matrix bases [62]. Specifically, the Pauli matrices $\{\mathbb{I}, X, Y, Z\}$ provide a convenient basis, allowing an arbitrary 2×2 unitary matrix A to be expressed as a linear combination of these basis matrices.

An arbitrary single-qubit operation matrix A can thus be decomposed into:

$$A = \frac{1}{2}\left(\mathrm{Tr}(A\mathbb{I})\,\mathbb{I} + \mathrm{Tr}(AX)\,X + \mathrm{Tr}(AY)\,Y + \mathrm{Tr}(AZ)\,Z\right), \tag{2}$$

where $\mathrm{Tr}(\cdot)$ denotes the trace operation. However, directly implementing this decomposition in quantum hardware is challenging due to the complex amplitudes that quantum systems inherently cannot measure directly.

To resolve this, the Pauli matrices are further expanded into their eigenbases, allowing the decomposition to be reorganized into a form compatible with measurement and state preparation operations on quantum hardware:

$$A = \frac{1}{2}\left(A_1 + A_2 + A_3 + A_4\right), \tag{3}$$

where each component A_i is defined explicitly as:

$$A_1 = \left[\mathrm{Tr}(A\mathbb{I}) + \mathrm{Tr}(AZ)\right]\langle 0|0\rangle, \tag{4}$$

$$A_2 = \left[\mathrm{Tr}(A\mathbb{I}) - \mathrm{Tr}(AZ)\right]\langle 1|1\rangle, \tag{5}$$

$$A_3 = \mathrm{Tr}(AX)\left[2\langle +|+\rangle - \langle 0|0\rangle - \langle 1|1\rangle\right], \tag{6}$$

$$A_4 = \mathrm{Tr}(AY)\left[2\langle +i| + i\rangle - \langle 0|0\rangle - \langle 1|1\rangle\right], \tag{7}$$

with eigenstates defined as:

$$\langle +| = \frac{1}{\sqrt{2}}(\langle 0| + \langle 1|), \tag{8}$$

$$\langle +i| = \frac{1}{\sqrt{2}}(\langle 0| + i\langle 1|). \tag{9}$$

Through this decomposition, quantum circuit cutting effectively transforms complex operations into measurable eigenstates and corresponding state preparations, thereby enabling the circuit to be segmented at designated qubits. This capability significantly enhances the feasibility of quantum computations by reducing circuit complexity and allowing execution on hardware with limited qubit coherence and connectivity.

2.3. Quantum and quantum-inspired models for multimodal tasks

The inherent quantum characteristics—such as superposition, entanglement, and interference—render quantum strategies instinctively suited for addressing the complexities of multimodal tasks. While many models incorporate quantum-inspired strategies to leverage these phenomena, the practical deployment of truly quantum models remains limited due to the significant challenges and hardware constraints associated with quantum computing.

Quantum-inspired models are fundamentally deep-learning based classical frameworks that emulate quantum-like strategies. These approaches draw upon the conceptual underpinnings of quantum mechanics to enhance traditional computational methods without requiring actual quantum hardware. For instance, the Quantum Fuzzy Neural Network (QFNN) discussed in [9] integrates quantum and fuzzy logic principles to improve sentiment and sarcasm detection in noisy environments. Similarly, the framework outlined in [10] employs quantum theoretical constructs to build robust multimodal representations for video sequences, thereby enhancing sentiment analysis based on textual queries. Such methods underscore the potential of quantum-inspired techniques to address the inherent intricacies and variabilities of multimodal data, offering marked improvements over conventional approaches.

In contrast, truly quantum models leverage genuine quantum computing mechanisms to process and analyze data—an approach that is particularly advantageous for managing large volumes of high-dimensional information. However, due to practical challenges, only a handful of these real quantum models exist. For example, the QNMF system in [11] deploys a quantum convolutional neural network (QCNN) combined with a variational quantum classifier (VQC) to efficiently handle and integrate multimodal medical data from the Internet of Medical Things (IoMT). This system not only achieves notable accuracy in diagnosing conditions like breast cancer and Covid-19 but also demonstrates robust resistance to quantum noise. These properties illustrate the transformative potential of genuine quantum computing in fields that demand speed, precision, and resilience.



Fig. 1. Sample visualization from the CMU-MOSI dataset.

As Table 1 shown, our proposed QD-MSA model is explicitly designed to thrive on current NISQ hardware: its low qubit footprint and modular circuit-splitting methods substantially mitigate decoherence and gate errors, resulting in a solution that is both more resilient to noise and scalable for near-term quantum deployments.

3. Datasets and notations

In this study, two benchmark datasets are utilized for multimodal sentiment analysis [63,64]: the Multimodal Corpus of Sentiment Intensity (CMU-MOSI) dataset and the CMU Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI) dataset.

3.1. CMU-MOSI dataset

The CMU-MOSI dataset [4] includes 2199 opinion video clips, each annotated with a sentiment intensity score ranging from −3 (strongly negative) to +3 (strongly positive). The dataset features comprehensive annotations covering subjectivity and sentiment intensity, along with detailed visual features per frame and per opinion, and audio features per millisecond. These rich annotations facilitate a detailed analysis of multimodal signals and their correlation with expressed sentiments (see Fig. 1).

3.2. CMU-MOSEI dataset

The CMU-MOSEI dataset [65] represents the largest available resource for multimodal sentiment analysis and emotion recognition. It encompasses over 23,500 sentence utterance videos sourced from more than 1000 YouTube contributors, ensuring a gender-balanced representation. The sentences, selected randomly from a diverse range of topics and monolog videos, ensure content variety. Each video is meticulously transcribed with accurate punctuation to support precise textual analysis (see Fig. 2).

The annotation process of CMU-MOSEI builds upon the methodologies used in CMU-MOSI and the Stanford Sentiment Treebank. Sentiment in each sentence is annotated on a Likert scale from −3 to +3, indicating a range from highly negative to highly positive. Additionally, emotions based on Ekman's model—happiness, sadness, anger, fear, disgust, and surprise—are annotated on a scale from 0 to 3, reflecting the intensity of each emotion from no evidence to a strong presence.

3.3. Dataset preparation and division

To ensure a rigorous evaluation framework, we employ a k-fold cross validation method that systematically rotates the training, testing, and validation splits across multiple iterations. Specifically, for the CMU-MOSI dataset, originally comprising 2183 samples, 17 samples were randomly duplicated to yield a total of 2200 samples. The resulting dataset was then partitioned into three disjoint subsets, allocating

Table 1
Comparative analysis of multimodal methods.

Category	Example	Params.	Hardware	Advantages	Limitations
Deep-learning Models	TFN [4], MARN [5]	25–70M	GPU (>16 GB)	High accuracy, well-established	Large models, long training times
Quantum-inspired Models	QFNN [9,10]	5–20M	GPU/CPU	Robust feature fusion, noise tolerance	Classical resources, no quantum speedup
Quantum Models	QNMF [11]	1–2M	NISQ Quantum Computer	Quantum resilience, exponential scaling potential	Limited coherence, errors, shallow circuits
Proposed QD-MSA	This work	<0.5M	NISQ Quantum Computer	Efficient fusion, minimal qubits, optimized for NISQ, enhanced error mitigation	Hardware-dependent, requires calibration

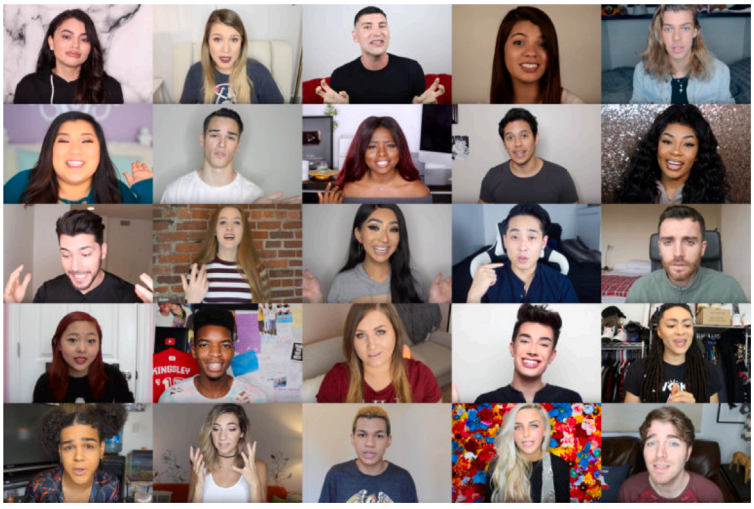


Fig. 2. Sample visualization from the CMU-MOSEI dataset.

60% for training, 30% for testing, and 10% for validation, as this is a relatively small dataset and we aim to maintain enough samples for accurate testing. In a similar fashion, the CMU-MOSEI dataset, consisting of 22,860 samples, was divided according to the same proportional scheme: 70% of the data was used for training, 20% for testing, and the remaining 10% for validation. This standardized approach to dataset preparation and division facilitates a comprehensive and balanced evaluation of model performance across all data subsets.

3.4. Notations

The notations used in this paper are summarized in Table 2.

4. Methodology

In this section, the proposed model for multimodal sentiment analysis is introduced. It is specifically designed to effectively integrate textual, audio, and visual data. The architecture of the model comprises two principal components: the feature extraction module and the quantum split convolution network module, which is responsible for feature fusion.

4.1. Overall workflow

Our proposed model operates through a two-part workflow: Feature Extraction and Feature Fusion and Prediction. This division leverages classical and quantum computing resources to enhance performance in multimodal sentiment analysis (see Fig. 3).

Feature Extraction (Classical Computing): In the feature extraction phase, raw video data is separated into three distinct modalities: visual, acoustic, and textual. Each modality undergoes preprocessing to remove noise and irrelevant information, ensuring that the subsequent feature extraction is based on high-quality data. The preprocessed data for each modality is then fed into Gated Recurrent Units (GRU) networks tailored specifically to capture temporal dependencies and extract modality-specific features. After extraction, the features from all modalities are concatenated into a unified feature vector. To prepare for the quantum phase, this high-dimensional vector is subsequently reduced in dimensionality using a Multi-Layer Perceptron (MLP), ensuring compatibility with quantum tensor networks.

Feature Fusion and Prediction (Quantum Computing): The feature fusion and prediction phase leverages quantum computing resources, potentially utilizing quantum computer clusters for distributed processing. The dimensionally reduced feature vector is first encoded into a quantum state using angle encoding, which involves applying Hadamard gates to create a superposition of basis states followed by RY gates that rotate each qubit based on the feature values. This encoded quantum state is then divided into smaller, independent circuits, which are inserted into a quantum program queue for concurrent processing by one or more quantum computers or clusters. The isolation of these split circuits allows them to be executed in parallel without interference, thereby mitigating errors and enhancing computational efficiency. Once processed, the outputs from the quantum circuits are collected and used to reconstruct probability distributions, ensuring accurate aggregation of results. These reconstructed probabilities are then passed through an MLP followed by a softmax layer to generate the final sentiment scores. These scores are instrumental in computing the loss, which in turn optimizes the quantum circuit parameters (ansatz) through backpropagation.

By applying this workflow, our model significantly enhances performance in multimodal sentiment analysis tasks. Compared to classical fusion methods, quantum fusion offers greater resilience and

Table 2
Notations used in the paper.

Type	Notation	Description
Input	u	Input discourse
	$w_1, w_2, \ldots, w_l$	Words in the input discourse
	l	Number of words
	X_{ij}	Number of times word i appears in context of word j
	X_i	Number of times word i appears in the corpus
	X_j	Number of times word j appears in the corpus
Output	GloVe(w_i)	GloVe embedding for word w_i
	h_t	Hidden state at time t
	z_t	Update gate at time t
	r_t	Reset gate at time t
	$\tilde{h}_t$	Candidate activation at time t
	$\mathbf{v}_{\text{text}}$	Resulting textual feature vector
	$\mathbf{h}_{\text{GRU}}$	Output of the GRU network
	$\mathbf{h}_t, \mathbf{h}_a, \mathbf{h}_v$	Feature vectors for text, audio, and visual modalities
	$\mathbf{h}_{\text{fused}}$	Unified feature vector
	MSE	Mean squared error function
	y_α	Classification outputs
	p_{uncut}	Probability of the uncut circuit
Quantum machine Learning	W_z, W_r, W	Learnable weight matrices
	W_{text}	Learnable weight matrix for text
	$\mathbf{b}_{\text{text}}$	Bias term for text
	$\mathcal{O}(\log(N))$	Logarithmic parameter scaling
	$\langle\psi_\alpha\|U_i V_j F\|\psi_\alpha\rangle$	Quantum circuit operation
	$\{\mathbb{I}, X, Y, Z\}$	Set of Pauli matrices
	A	Arbitrary 2×2 unitary matrix
	$\text{Tr}(\cdot)$	Trace operation
	A_1, A_2, A_3, A_4	Terms in the eigenbasis decomposition
	$\langle 0\|, \langle 1\|, \langle +\|, \langle +i\|$	Quantum states

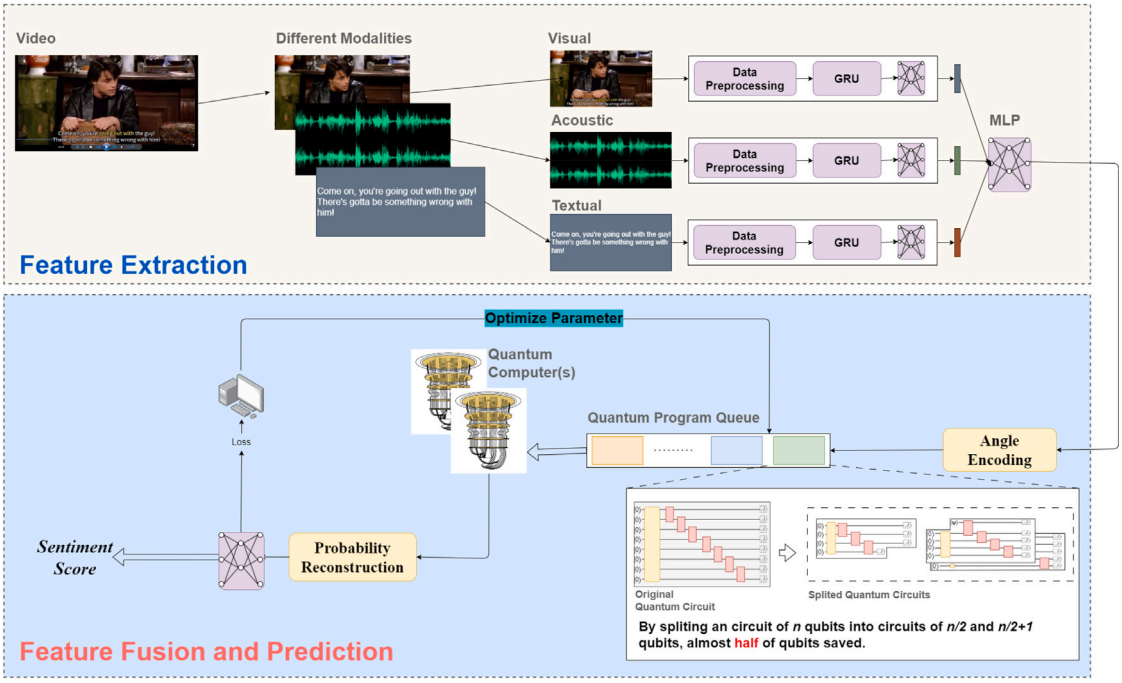


Fig. 3. Overall workflow of our multimodal sentiment analysis method.

efficiency by fully utilizing quantum system characteristics such as superposition and entanglement while reducing the number of required parameters. Additionally, quantum tensor networks excel at extracting high-dimensional features, enabling the discovery of intricate hidden patterns that classical methods may miss. The distributed architecture of our approach allows the model to scale efficiently, handling more complex tasks by distributing circuit execution across quantum computer clusters. This integration of classical feature extraction with quantum feature fusion and prediction leverages the strengths of both computing paradigms, resulting in a robust and efficient multimodal sentiment analysis system.

4.2. Model overview

The model comprises a feature extraction module that processes raw data from each modality to extract pertinent features. Specifically, textual (h_t), visual (h_a), and acoustic (h_v) data are separately sent to

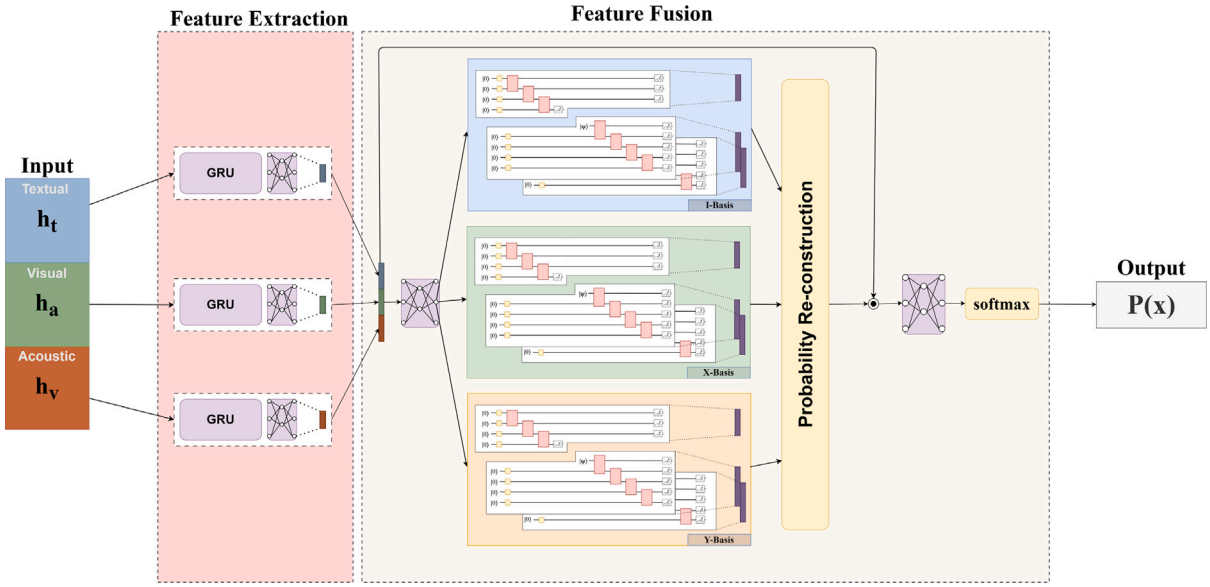


Fig. 4. Overview of the proposed multimodal sentiment analysis model.

GRU-based feature extraction heads, which are implemented to effectively handle sequenced and aligned data characteristics. This approach leverages the strengths of GRUs in capturing temporal dependencies within each modality.

After feature extraction, the three modalities are concatenated and passed to the feature fusion module. In this module, a quantum tensor network is utilized as the feature fusion mechanism, fully exploiting the entanglement and superposition characteristics of quantum systems while using fewer parameters compared to traditional methods. To further optimize resource usage, a circuit splitting method is applied within the quantum tensor network, nearly halving the qubit usage and enabling the execution of more complex quantum circuits on limited-qubit quantum computers.

Within the feature fusion module, the concatenated features are first reduced in dimensionality via a multilayer perceptron (MLP). These reduced features are then quantum encoded and processed using the quantum tensor network. The results of quantum measurements are calculated to obtain accurate data representations, which are subsequently combined with the concatenated features from before the fusion module. This combined data is further processed by another MLP and a final softmax layer to produce the correct classification results.

The comprehensive structure of the model is depicted in Fig. 4, illustrating the separate processing of different modalities and the integration of the quantum tensor network for feature fusion.

4.3. Data preprocessing and feature extraction

To effectively leverage multimodal information, the data undergo comprehensive preprocessing and modality-specific feature extraction.

4.3.1. Data preprocessing

Prior to feature extraction, multimodal data alignment is essential to ensure that audio, visual, and textual modalities correspond at the word-level. Given an input discourse u comprising words $w_1, w_2, \ldots, w_l$, where l represents the total number of words, the Python library GENTLE is utilized for forced alignment. This process synchronizes audio signals precisely with individual words, enabling subsequent extraction of word-aligned multimodal features from all modalities.

For textual preprocessing, linguistic data are converted into semantic vectors through GloVe embeddings [66]. For a given word w_i, the GloVe embedding is computed as:

$$\text{GloVe}(w_i) = \log\left(\frac{X_{ij}}{X_i X_j}\right) = \log(X_{ij}) - \log(X_i) - \log(X_j), \tag{10}$$

where X_{ij} is the co-occurrence frequency of words i and j, and X_i, X_j are the frequencies of words i and j respectively in the corpus.

For visual preprocessing, video frames are processed at 30 frames per second. Facial features, including 68 facial landmarks, 20 facial shape parameters, Histogram of Oriented Gradients (HoG), head pose, orientation, eye gaze, and facial action units (FACS) [67], are extracted using OpenFace and MTCNN [68]. Basic emotions and face embeddings generated by DeepFace [69] further enrich the visual representation.

For acoustic preprocessing, COVAREP [70] software extracts essential vocal characteristics including Mel-frequency cepstral coefficients (MFCCs), pitch features, voiced/unvoiced segments, glottal source parameters, peak slope parameters, and maxima dispersion quotients.

4.3.2. Feature extraction via modality-specific encoders

After alignment and preprocessing, each modality undergoes specialized encoding to extract distinct characteristics and temporal dependencies.

The encoding process in each modality utilizes a Gated Recurrent Unit (GRU) network, which is designed to model sequential dependencies effectively. The GRU operates with gating mechanisms defined as:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t]), \tag{11}$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t]), \tag{12}$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t]), \tag{13}$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t, \tag{14}$$

where x_t is the input vector at time step t, h_{t-1} is the previous hidden state, z_t and r_t are the update and reset gates, and W_z, W_r, and W are learnable weights.

For the textual modality, the semantic embeddings from GloVe are input to a GRU, producing contextualized word-level representations. These outputs are passed through a linear transformation to yield the textual feature vector:

$$\mathbf{v}_{\text{text}} = W_{\text{text}} \cdot \mathbf{h}_{\text{GRU}} + \mathbf{b}_{\text{text}}. \tag{15}$$

For the visual modality, preprocessed visual features are encoded using a GRU to capture temporal dependencies in facial expressions and head movements. The resulting sequence representation is projected through a linear layer for dimensionality reduction.

For the acoustic modality, GRU encoding is also applied to the time-series acoustic features to model vocal signal dynamics. A linear layer maps the GRU outputs to a fixed-size acoustic vector representation.

4.3.3. Unified feature representation

The final outputs from the modality-specific encoders—the textual feature vector $\mathbf{h}_t \in \mathbb{R}^{600}$, acoustic feature vector $\mathbf{h}_a \in \mathbb{R}^{200}$, and visual feature vector $\mathbf{h}_v \in \mathbb{R}^{200}$—are concatenated into a unified multimodal feature vector:

$$\mathbf{h}_{\text{fused}} = [\mathbf{h}_t; \mathbf{h}_a; \mathbf{h}_v] \in \mathbb{R}^{1000}. \tag{16}$$

This combined representation encapsulates rich semantic, acoustic, and visual cues, offering a robust basis for the subsequent fusion and emotion recognition tasks.

4.4. Quantum split tensor network module for feature fusion

4.4.1. Quantum encoding

The Quantum Encoding module transforms the unified feature representation into a quantum state, enabling efficient multimodal data processing. This encoding process applies a quantum gate U to the unified feature vector, resulting in the quantum state $|\psi\rangle$.

We employ angle encoding, which involves applying Hadamard gates followed by RY gates on each qubit. First, Hadamard gates create a superposition of all basis states, and then RY gates rotate each qubit by specific angles to encode the data. This method ensures that the quantum state accurately captures the input data's characteristics, facilitating precise and robust quantum computations.

4.4.2. Quantum tensor network architecture

A tensor network is a collection of tensors where a subset of all indices is contracted. In our Quantum Tensor Network (QTN) architecture, designed for classifying N-qubit input states, we leverage this framework to manage and process quantum information efficiently. Specifically, we employ eight qubits arranged in a Matrix Product State (MPS) configuration as the backbone of our model. This MPS arrangement facilitates efficient entanglement and information sharing across qubits, fully utilizing the quantum system's inherent characteristics such as superposition and entanglement while minimizing the number of required parameters.

A quantum tensor network extends the concept of tensor networks to quantum circuits. One approach to making tensor-network quantum circuits compatible with quantum hardware is to replace the unitary blocks with local circuits [71]. In this context, tensor-network quantum circuits can be viewed as strategies for structuring smaller circuit ansatzes, effectively serving as meta-ansatzes. This allows the use of established techniques for designing, implementing, and optimizing variational quantum circuits. By using tensor networks as templates to generate tensor-network meta-ansatzes and replacing the tensor blocks with parameterized circuits, we obtain a final variational quantum circuit that can be simulated, implemented on hardware, and optimized akin to other variational circuits.

The motivation for selecting a quantum tensor network as our fusion backbone lies in its ability to fully utilize the quantum system's characteristics—specifically superposition and entanglement—while reducing the number of parameters. Moreover, compared to Quantum Convolutional Neural Networks (QCNN), quantum tensor networks are more adept at extracting high-dimensional features, enhancing the network's capacity to process complex multimodal data efficiently.

The QTN employs a mean squared error (MSE) loss function during the learning process, defined as:

$$\text{MSE} = \frac{1}{2M} \sum_{\alpha=1}^{M} \left(y_\alpha - \langle \psi_\alpha | U_i V_j F | \psi_\alpha \rangle \right)^2 \tag{17}$$

where $|\psi_\alpha\rangle$ are the input states, y_α are classification outputs, and U_i, V_j, F represent the variational parameters of the quantum circuit.

Tensor networks, comprising multi-dimensional arrays that extend the concepts of vectors and matrices, serve as the fundamental framework for organizing quantum information flow and transformation.

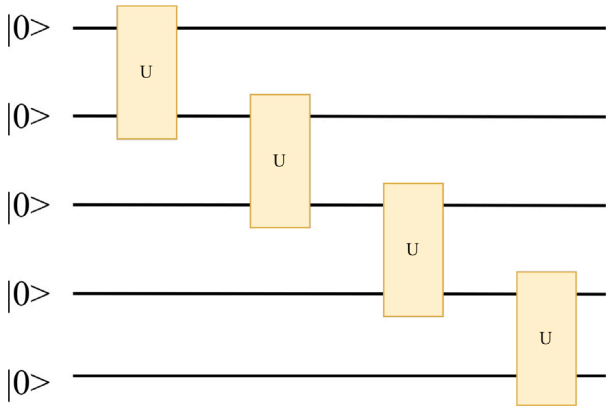


Fig. 5. Matrix Product State (MPS) structure used in the convolution layers of the QTN.

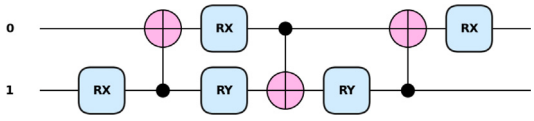


Fig. 6. Illustration of the tensor block.

Specifically, Matrix Product States (MPS) and Tree Tensor Networks (TTN) [71] are applied to structure the convolutional and pooling layers. In our model, an MPS configuration, illustrated in Fig. 5, is employed within the quantum circuit to facilitate efficient entanglement and information sharing across qubits.

Compared to traditional quantum convolutional networks, our architecture fully utilizes information across all qubits without losing bit width. This comprehensive utilization enables more effective processing and classification of quantum information.

The fundamental unit of our network is the tensor block, depicted in Fig. 6. This design enhances the network's sensitivity and facilitates more effective learning. Each tensor block comprises RX, RY, and CNOT gates, optimized for feature extraction. By exclusively utilizing tensor blocks as the backbone, our network optimizes its capability to extract and process quantum features efficiently. In our implementation, we utilize the network in a 8-qubit architecture, ensuring optimal performance.

4.4.3. Circuit cutting implementation

In the proposed implementation, the circuit cutting method we used in Section 2.2.2 is applied to the quantum tensor network by splitting the circuit between qubits 4 and 5. This partition divides the original circuit into two subcircuits, which can be executed separately on quantum hardware with fewer qubits. The subcircuits are denoted as subcircuit_1 and subcircuit_2. The qubit at the cut location (qubit 4 in subcircuit_1 and qubit 5 in subcircuit_2) is treated according to the decomposition in Eq. (3). Specifically, measurements are performed on qubit 4 in the $\mathbb{I}, X, Y$ bases (since measuring circuits in the $\mathbb{I}$ and Z have the same physical implementation), and qubit 5 is prepared in the corresponding eigenstates $|0\rangle, |1\rangle, |+\rangle, |+i\rangle$ in subcircuit_2.

The outputs of the subcircuits are represented using the notation:

$$C(\langle q_0, q_1, \ldots, q_{n-1}|; M_0, M_1, \ldots, M_{n-1}), \tag{18}$$

where C is the circuit, $\langle q_i|$ are the initial states of the qubits, and M_i are the measurement bases for each qubit.

4.4.4. Reconstruction of original circuit

To reconstruct the output of the original uncut circuit, the measurement results of the subcircuits are classically combined. For each

Table 3
Hyperparameters used in the experiments.

Hyperparameter	Value
Dropout	1×10^{-1}
Learning Rate (lr)	1×10^{-4}
Learning Rate Optimization	Step LR, $\gamma = 0.5$, step_size = 5
Optimizer	Adam
Weight Decay	1×10^{-3}
Hidden Layers (MLP)	512
Batch Size	64
Qubit usage for QTN Splited module	4 (upper part) and 5 (lower part)
Qubit usage for QTN Unsplited module	8

term in the decomposition, the probabilities from the subcircuits are calculated, and Kronecker products are performed, summing over all terms:

$$p_{\text{uncut}} = \frac{1}{2} \sum_{i=1}^{4} p_{1,i} \otimes p_{2,i}, \tag{19}$$

where $p_{1,i}$ and $p_{2,i}$ are the probabilities obtained from subcircuit$_1$ and subcircuit$_2$, respectively, corresponding to the ith term in the decomposition.

4.4.5. Workflow of the quantum split tensor network architecture

The workflow for implementing the Quantum Split Tensor Network Architecture begins with circuit partitioning. First, the qubit(s) where the circuit will be split are identified; in this case, the split occurs between qubits 4 and 5. The original circuit is then divided into two subcircuits, subcircuit$_1$ and subcircuit$_2$, based on these selected cut points.

Next, each subcircuit is executed independently. In subcircuit$_1$, qubit 4 is measured in the $\mathbb{I}, X, Y, Z$ bases, while in subcircuit$_2$, qubit 5 is prepared in the corresponding eigenstates $|0\rangle, |1\rangle, |+\rangle, |+i\rangle$ based on the measurement outcomes from subcircuit$_1$. These subcircuits run on quantum hardware with a reduced number of qubits, facilitating parallel execution and mitigating errors.

Finally, the probability distributions from the subcircuits are reconstructed classically. For each term in the decomposition, probabilities from subcircuit$_1$ and subcircuit$_2$ are calculated and combined using Kronecker products. These combined probabilities are then summed according to Eq. (19) to accurately reconstruct the probability distribution of the original uncut circuit, effectively restoring the full quantum computation through classical post-processing.

5. Experimental setup

We use PennyLane version 0.38.0 along with the PennyLane-lightning-gpu simulator as our quantum devices, and PyTorch 2.3.0 as the primary framework for our learning tasks. The main server used for our experiments is equipped with an Intel Xeon Platinum 8352 V processor (144 cores) operating at 3.5 GHz, coupled with four NVIDIA GeForce RTX 4090 GPUs to ensure efficient execution of both classical and quantum computations. These hardware resources were chosen to provide the necessary computational power for our experiments, allowing us to efficiently run complex models and simulations.

To evaluate the performance of our proposed model, we conducted a series of experiments with meticulously selected hyperparameters. Table 3 provides a comprehensive summary of the hyperparameters employed in these experiments. These parameters were carefully optimized to enhance model performance, thereby contributing to the superior results observed in our study.

6. Results analysis

6.1. Performance on multimodal sentiment analysis

The performance evaluation of various models for multimodal sentiment analysis in this study includes both deep learning models and models using quantum-inspired strategies. Deep learning models, such as EF-LSTM [5], LF-LSTM [5], TFN [4], MARN [5], MFN [5], and MuLT [72], have been widely used for sentiment analysis tasks, and they demonstrate considerable effectiveness in capturing sentiment from multimodal data. These models rely on sophisticated architectures that combine multiple modalities like text, audio, and video to predict sentiment. On the other hand, the quantum-inspired strategy, exemplified by the QMF model, explores the integration of quantum computing principles into deep learning models, though quantum computing hardware limitations prevent the use of fully quantum-based models in this domain. Due to the lack of fully quantum models for multimodal sentiment analysis, only one quantum-inspired model, QMF [9], is included for comparison in this study. As such, the proposed QD-MSA model, which builds upon deep learning strategies while incorporating elements of quantum-inspired techniques, is tested and compared against these established models.

The following Tables 4 and 5 present the performance of various models on the CMU-MOSEI and CMU-MOSI datasets. The bolded values indicate the best performance metrics achieved by each model.

From Table 4, the QD-MSA model achieves the highest accuracy in both the Acc-7 (0.4811) and Acc-2 (0.8162) metrics, outperforming other models, including QMF. It also records the best F1 score (0.8251) and the lowest Mean Absolute Error (MAE) of 0.5069. While MARN attains a higher correlation coefficient (Corr) of 0.6515, the QD-MSA model demonstrates competitive performance across all metrics, highlighting its effectiveness in leveraging multimodal features for sentiment analysis on the CMU-MOSEI dataset.

In Table 5, the QD-MSA model achieves the highest Acc-7 (0.3394) on the CMU-MOSI dataset. While the MuLT model leads in both Acc-2 (0.8002) and F1 score (0.7980), the proposed model remains competitive with an Acc-2 of 0.7780 and an F1 score of 0.7816. Additionally, it achieves the best MAE of 0.9025 and a correlation coefficient of 0.6816, closely trailing QMF's 0.6909. These results demonstrate the model's ability to effectively capture sentiment variations across different metrics on the CMU-MOSI dataset.

Since our model's performance on CMU-MOSEI is stronger than on CMU-MOSI, we designed an experiment to investigate the impact of varying training set sizes on our QD-MSA model. We altered the size of the training set while keeping other experimental parameters constant. Fig. 7 illustrates the change in QD-MSA's performance metrics as the training set proportion of the CMU-MOSEI dataset is varied from 1% to 70%.

From the experimental results shown in Fig. 7, we observe that the performance of our model is relatively poor with smaller training set sizes. This observation helps explain why the results on the CMU-MOSI dataset (which has a size of approximately 2000 samples) are comparatively lower than those achieved by other deep learning models. QD-MSA does not appear to excel at learning features from

Table 4
Performance of different models on the CMU-MOSEI dataset.

Model	Acc-7	Acc-2	F1	MAE	Corr
Deep Learning Models					
EF-LSTM [5]	0.4621	0.7733	0.7739	0.6315	0.5957
LF-LSTM [5]	0.4557	0.7718	0.7663	0.6314	0.5664
TFN [4]	0.4793	0.8018	0.8019	0.6465	0.6511
MARN [5]	0.4742	0.7823	0.7856	0.6270	0.6515
MFN [5]	0.4084	0.7797	0.7525	0.6349	**0.6562**
MuLT [72]	0.4322	0.7928	0.7847	0.6315	0.6411
Model using Quantum-Inspired Strategy					
QMF [9]	0.4687	0.7612	0.7744	0.6273	0.6446
Our Model					
QD-MSA	**0.4811**	**0.8162**	**0.8251**	**0.5069**	0.5935

Table 5
Performance of different models on the CMU-MOSI dataset.

Model	Acc-7	Acc-2	F1	MAE	Corr
Deep Learning Models					
EF-LSTM [5]	0.3123	0.7230	0.7261	1.1110	0.5584
LF-LSTM [5]	0.2978	0.7334	0.7360	1.0966	0.6008
TFN [4]	0.3115	0.7397	0.7344	0.9705	0.6277
MARN [5]	0.2994	0.7580	0.7723	0.9207	0.6542
MFN	0.2889	0.7773	0.7706	0.9222	0.5907
MuLT [72]	0.3372	**0.8002**	**0.7980**	**0.8898**	0.6869
Model using Quantum-Inspired Strategy					
QMF [9]	0.3076	0.7950	0.7946	0.9146	**0.6909**
Our Model					
QD-MSA	**0.3394**	0.7780	0.7816	0.9025	0.6816

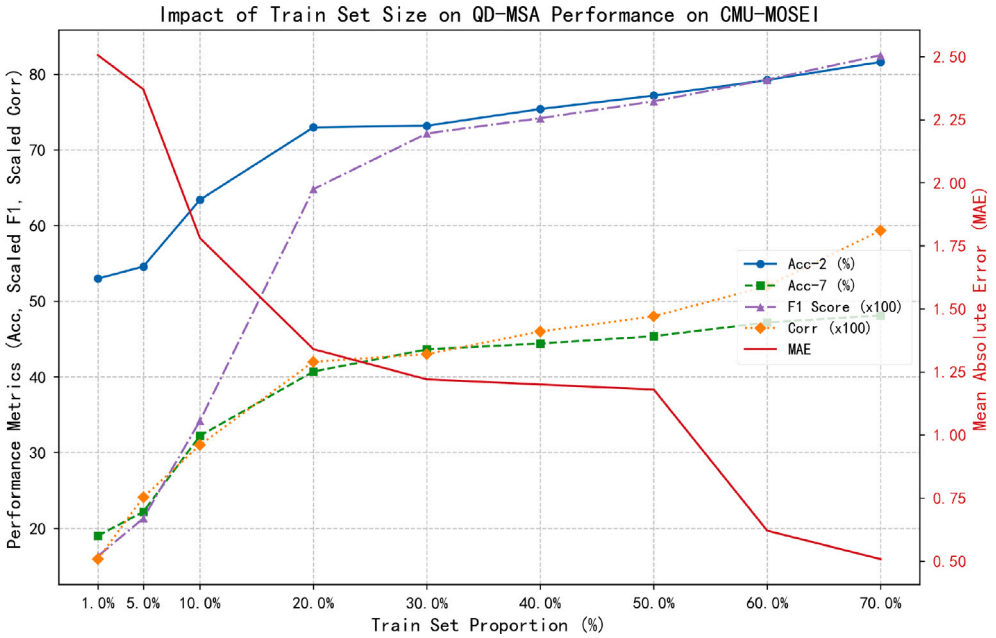


Fig. 7. Impact of train set size on QD-MSA performance on CMU-MOSEI.

smaller sample sizes, and its stability is notably reduced when trained on subsets of the CMU-MOSEI dataset where the training set size is less than 20% (approximately 4000 samples).

Analyzing the network structure, several factors likely contribute to our model's less favorable performance on small datasets compared to deep learning models. Firstly, the optimizer we used (Adam) is not specifically designed for optimizing quantum models. Given the relatively small number of layers and parameters in our model, this mismatch in optimization can lead to training instability. Secondly, our model heavily relies on quantum rotation operations. These operations are coupled with each other, making the parameter updates for these rotations require multiple adjustments to reach an optimal solution. Furthermore, the angles of the quantum rotation operations are typically small. This characteristic makes it easier for a poorly trained model to produce final classification results that fall into an intermediate category rather than clearly separating classes.

Table 6
Ablation study comparing different model architectures.

Fusion Model	Acc-7	Acc-2	F1	MAE	Corr	Inference Time(s)	Params of Quantum Layer	Total Params of Fusion Model
MLP	0.4202	0.7808	0.7810	0.5172	0.4840	2.31	*NaN*	516 103
QTN Unsplited	0.4972	0.8203	0.8310	0.4927	0.6184	42.13	42	7357
QTN Splited	0.4811	0.8162	0.8251	0.5069	0.5935	121.94	126	7357

Table 7
Unimodal and bimodal sentiment classification results on CMU-MOSEI.

Modal	Acc-7	Acc-2	F1	MAE	Corr
Vision	0.4093	0.7793	0.7609	0.7644	0.4094
Audio	0.3919	0.7640	0.7427	0.4893	0.3917
Text	0.4571	0.8475	0.8058	0.5432	0.4568
Vision + Audio	0.4018	0.7780	0.7685	0.5985	0.4015
Audio + Text	0.4706	0.8304	0.8015	0.5293	0.4707
Vision + Text	0.4524	**0.8328**	0.4451	0.5816	0.4184
Vision + Audio + Text	**0.4811**	0.8162	**0.8251**	**0.5069**	**0.5935**

6.2. Ablation study

To assess the effectiveness of the proposed feature fusion module, we maintain consistent feature extraction across all models and vary only the feature fusion strategy. This ablation study compares the QTN(quantum tensor network) splited fusion module with QTN unsplited and a classical multilayer perceptron (MLP).

Table 6 presents the comparison results. The classical MLP model serves as a baseline, utilizing 516,103 parameters. In contrast, both QTN fusion models employ significantly fewer parameters, with the QTN Unsplited and QTN Splited models using only 7357 total parameters.

The QTN Unsplited model outperforms the MLP in most metrics, achieving higher Acc-7 (0.4972), Acc-2 (0.8203), F1 score (0.8310), lower MAE (0.4927) and Corr(0.6184). Similarly, the QTN splited model demonstrates competitive performance with Acc-7 of 0.4811 and Acc-2 of 0.8162, while maintaining the same total number of parameters as the unsplit version.

Importantly, both quantum fusion approaches significantly reduce the number of parameters compared to the classical MLP while either matching or exceeding its performance. This highlights the parameter efficiency of our quantum tensor-based fusion modules, demonstrating that effective multimodal feature integration can be achieved with substantially fewer resources.

The second set of experiments focuses on the contribution of each modality to the overall performance. The model is evaluated under various modality settings, including unimodal (vision, audio, or text) and bimodal (vision + audio, audio + text, or vision + text) configurations. The results are summarized in Table 7. It is evident that even with limited modalities, the proposed network maintains a satisfactory performance. Notably, the model achieves the best results when utilizing the text modality alone, which can be attributed to the distribution of the source data.

These ablation studies highlight the effectiveness of the proposed Quantum Split Tensor Network Architecture in reducing the required number of qubits while maintaining competitive performance. Furthermore, the experiments demonstrate the model's robustness under different modality settings, with the text modality contributing the most to the overall performance in the given dataset.

7. Conclusion and future work

In this work, we propose QD-MSA, a novel quantum multimodal sentiment analysis model that leverages quantum tensor networks to effectively fuse diverse modalities. Our experimental results on the CMU-MOSI and CMU-MOSEI datasets demonstrate that QD-MSA outperforms state-of-the-art deep-learning-based models and classical approaches relying solely on quantum-inspired strategies—all while utilizing a fraction of the parameters required for conventional modality fusion. Notably, by introducing quantum circuit splitting techniques to the multimodal sentiment analysis task, we achieve nearly a 50% reduction in qubit usage, enabling the execution of larger quantum circuits on devices with limited qubit resources. In addition, our distributed workflow, designed specifically for quantum computer clusters, significantly accelerates the model and mitigates errors typical in the Noisy Intermediate-Scale Quantum (NISQ) era.

This work represents a significant breakthrough in the practical application of quantum computing within the NISQ era. By integrating a quantum-hybrid model with an advanced distributed workflow, we enable effective quantum computation on hardware constrained by limited qubits and inherent noise. Our approach facilitates complex tasks such as video analysis and human sentiment classification, achieving performance that rivals or even surpasses that of traditional deep learning methods while requiring only 1% of the conventional parameter count in modal fusion. These advancements pave the way for more efficient, scalable, and robust applications of quantum computing in multimodal sentiment analysis and related fields.

Despite these advancements, our work has certain limitations that offer opportunities for future improvement. Firstly, our method incorporates quantum models only at the multimodal fusion stage, relying on classical deep learning models for feature extraction from individual modalities. This reliance means we may not be fully leveraging the potential advantages of quantum computing in processing and representing modal features. There is room to explore quantum-based approaches for feature extraction, which could enhance the overall performance and efficiency of the model by utilizing quantum properties more extensively throughout the pipeline. Secondly, due to the computational constraints of current quantum simulators and hardware, we employed a relatively simple quantum model with a limited number of qubits (eight qubits). This limitation restricts the model's capacity and expressiveness, potentially impeding its ability to capture more complex interactions between modalities. Furthermore, our experimental results indicate that QD-MSA exhibits less favorable performance on smaller datasets compared to established deep learning models. This suggests that the model may not be as effective at learning from limited samples, highlighting a potential area for improvement in its design or training methodology to enhance performance on datasets with fewer instances. Addressing these limitations will be crucial for unlocking the full potential of quantum computing in multimodal sentiment analysis and related fields.

Building on the current achievements in multimodal sentiment analysis, our future work will first focus on enhancing the text-modality processing capability of the quantum model. Inspired by the success of Transformer architectures in capturing long-range contextual dependencies, we aim to integrate quantum-adapted Transformer mechanisms into the hybrid framework. This extension is expected to refine

the extraction of semantic and syntactic features from text, potentially unlocking higher accuracy in complex sentiment reasoning tasks.

In parallel, we plan to generalize our model to broader multimodal scenarios beyond sentiment analysis, such as medical diagnosis and intent recognition. The demonstrated superiority of quantum-driven fusion in handling cross-modal interactions (e.g., audio-visual-text alignment in CMU-MOSEI) motivates this direction. However, new challenges may arise when integrating electrophysiological modalities like electroglottography (EGG): the distinct time-frequency characteristics of EGG signals, combined with their weak semantic correlation to other modalities, will necessitate optimizations to our quantum fusion layer — potentially through dynamic quantum gate parameterization or modality-specific subspace embedding.

Furthermore, to fully exploit quantum advantages in pre-processing, we propose developing a quantum-native feature extraction pipeline. In contrast to the current hybrid approach (classical CNNs/GRU for feature extraction followed by quantum fusion), this initiative will involve designing a "quantum dataloader" capable of directly encoding raw multimodal data (text spectrograms, EEG waveforms, etc.) into entangled quantum states. Subsequent parameterized quantum circuits will then extract high-dimensional features, reducing reliance on classical deep learning backbones while improving generalization across modalities.

Finally, we intend to implement the entire framework on real quantum hardware. By refining quantum circuit compilation strategies and adapting error mitigation protocols for NISQ devices, this effort will validate the scalability of our approach and lay the groundwork for practical quantum-AI integration in the NISQ computing era.

CRediT authorship contribution statement

Yuelin Li: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Yangyang Li:** Investigation, Funding acquisition, Formal analysis. **Zhengya Qi:** Validation, Software, Formal analysis. **Haorui Yang:** Validation, Software, Formal analysis. **Ronghua Shang:** Writing – review & editing, Investigation. **Licheng Jiao:** Writing – review & editing, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant Nos. 62476209 and 62176200, in part by the Key Research and Development Program of Shaanxi under Grant No. 2024CY2-GJHX-18, and in part by the Natural Science Basic Research Program of Shaanxi under Grant No. 2022JC-45.

Data availability

Data will be made available on request.

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